JOSEPH WU

SENIOR SOFTWARE ENGINEER | AI ENGINEERING

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PROFESSIONAL SUMMARY

Senior Software Engineer with 14 years of experience building backend platforms, distributed services, and AI-enabled applications across financial technology and product engineering. Python runs through that work: API design, data processing, cloud deployment, and the production operations that keep a system reliable after it ships.

Currently a Senior Full Stack Engineer at Arbitexia Limited, building AI-powered web features end to end with React, TypeScript, and Node.js. Integrates LLMs into product workflows, evaluates outputs against real user scenarios, and adds monitoring and fallbacks so AI features stay reliable. Also designs retrieval-augmented generation (RAG) pipelines, FastAPI services for document ingestion and semantic retrieval, and agent workflows, with evaluation around prompts and search so answers stay grounded in source material.

Previously built financial technology services at Wealthsimple and Wave HQ, including asynchronous processing, data integrations, PostgreSQL optimization, and internal tools. Reviews designs, plans releases with stakeholders, and mentors engineers on testing, deployment, and operational troubleshooting.

TECHNICAL EXPERTISE

Backend & Languages: Python, Java, Node.js, TypeScript, JavaScript, FastAPI, Django, Flask, REST APIs, SQL

AI & Data: LLM APIs, RAG, LangChain, LangGraph, embeddings, pgvector, semantic retrieval, agent workflows, prompt and output evaluation, PostgreSQL, Prisma, Redis

Cloud & Delivery: AWS, Docker, Kubernetes, Terraform, CI/CD, asynchronous processing, monitoring and observability, production operations

Frontend: React, Next.js, HTML, CSS, Tailwind CSS, responsive UI, accessibility

PROFESSIONAL EXPERIENCE

ARBITEXIA LIMITED | Senior Full Stack Engineer

Jan 2025 - Present | Florida, United States

- Built AI-powered web features end to end using React, TypeScript, and Node.js, from the user interface to the APIs that connected it to language models.
- Integrated LLMs into product workflows and evaluated outputs against real user scenarios to catch inconsistent or incorrect responses.
- Improved the reliability of AI features by tracking response times, API errors, and model usage, and adding fallbacks for failed requests.
- Helped shape technical decisions, reviewed code, and worked with product and design to bring features from prototype to production.

WEALTHSIMPLE | Senior Software Engineer

Apr 2019 - Dec 2024 | Delaware, United States

- Built backend services and internal tools for financial data processing, integrations, and operations at Wealthsimple US.
- Designed asynchronous workflows with retries and idempotency so failed jobs could be retried safely, and added monitoring to help diagnose production issues.
- Tuned PostgreSQL queries and caching, expanded automated test coverage, and improved release checks.
- Built React and TypeScript tools that helped operations teams track workflow status and investigate failures.
- Partnered with product, security, and infrastructure teams on system design and production rollouts; mentored engineers through code reviews and pair programming.

WAVE HQ | Software Engineer

Jul 2015 - Mar 2019 | Toronto, Ontario, Canada

- Built and maintained Python and Django services and REST APIs for small-business financial application workflows.
- Implemented background jobs, external API integrations, and data synchronization with clear failure handling.
- Improved SQL access patterns, automated test coverage, and logging while collaborating with product and frontend teams.

WAVE HQ | Junior Software Engineer

Jul 2012 - Jun 2015 | Toronto, Ontario, Canada

- Delivered application features using Python, Django, JavaScript, and relational databases under senior engineering guidance.
- Investigated defects, implemented validation and tests, and supported routine deployments and third-party integrations.

SELECTED ENGINEERING WORK

AI Knowledge Assistant: Retrieval-Augmented Generation. Designed a RAG architecture covering document ingestion, chunking, vector search, retrieval APIs, and source-grounded answers. Connected retrieval results to generated responses to keep outputs tied to supporting documents.

Workflow Automation: LLM Tool Calling. Prototyped a tool-calling workflow with deterministic validation checks and explicit approval steps before consequential actions, emphasizing predictable behavior and human oversight.

EDUCATION

UNIVERSITY OF WATERLOO | Bachelor's degree, Computer Science

2008 - 2012 | Waterloo, Ontario, Canada